



Synthetic Price Path Generation for European Gas Markets Using Generative Adversarial Networks with Applications to Option Pricing

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Abstract

Here goes abstract.

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1 Introduction

1.1 Context

1.2 Problem Statement and Motivation

1.3 Goals

1.4 Own Contribution

2 Financial Fundamentals

This section provides the financial background for the remainder of the thesis. It begins with general concepts of financial markets, returns and risk, and contrasts equity with commodities as asset classes. The section then focuses on the European natural gas market and the TTF hub. The futures and option contracts traded on it are introduced next, together with implied volatility. Finally, the stylized facts of commodity returns relevant for this thesis are described.

2.1 Financial Markets

A financial market is an organized venue in which participants trade financial assets. A financial asset is a claim to future value. An exchange is a market that operates under standardized rules, which specify what may be traded and how.

Buyers need to find sellers that are willing to sell the asset at a low price, while sellers need to find buyers that are willing to pay a high price. The bid is the highest price at which a participant is willing to buy an asset, and the ask is the lowest price at which a participant is willing to sell it. A transaction occurs when the two sides agree on a price. As a result, the observed price of an asset is not a fixed property but rather the outcome of the most recent transaction. A market is considered liquid if many participants can buy or sell continuously, without affecting the asset price (Harris 2003, pp. 3–7).

The return of an asset measures the relative change in its price over a given time period. The risk of an asset is the uncertainty of its return. It is generally measured with the standard deviation of its return, which is commonly referred to as volatility. The risk-free rate is the return of an investment that carries no risk, such as a short-term government bond. The Sharpe ratio is a measure of the risk-adjusted return of an asset

$$SR = \frac{\mu - r_f}{\sigma}, \quad (1)$$

where μ is the mean return over the period, r_f is the risk-free rate and σ is the standard deviation of the returns. SR is commonly annualized.

Hull (2022, p. 33) differentiates three main categories of traders. Hedgers trade to reduce the risk they are exposed to from future price movements. Speculators trade to take a position on the future direction of the price. Lastly, arbitrageurs combine long and short positions in two or more instruments, so that a profit is guaranteed regardless of the price movements. A participant who holds a long position benefits from a price increase, and a participant who holds a short position benefits from a price decrease.

Assets are grouped into asset classes according to their characteristics. Equity and commodities are two asset classes that are relevant for this thesis. **Equity** represents ownership in a company. The holder of equity is entitled to a share of the company's profits. The value of equity therefore depends on the company's future profits. Shares are traded on stock exchanges, financial markets that specialize in equity. **Commodities** are physical goods that are produced and consumed. The supply of a commodity is

determined by production and by the inventories that are held. Demand is generally inelastic, since the good is essential and has few substitutes in the short run. As a result, commodity prices are sensitive to physical conditions such as weather, storage levels and available transport capacity.

As stated above, equity and commodities differ in how their prices are formed. They also differ in what a trade eventually leads to. A trade in equity is settled in money, while a trade in a commodity can end in the delivery of the physical good, which the buyer has to receive and store. Lastly, the two differ in what their holders are exposed to. A holder of equity is affected only by the price of the shares, whereas a gas producer is affected by the price of gas and also by how much of it is sold, which changes with the weather and with economic conditions (Geman 2006). The European natural gas market is the commodity market studied in this thesis.

2.2 European Natural Gas Markets

Natural gas is a physical commodity that exists within a gas grid. The Title Transfer Facility (TTF) is a virtual trading hub in the Netherlands. A virtual trading hub allows participants to transfer ownership without reference to a specific physical location. TTF is operated by Gasunie Transport Services (GTS) within the Dutch gas transmission network. Gas futures contracts referencing TTF are traded on ICE Endex, the European energy exchange operated by Intercontinental Exchange. Futures contracts are described in Sec. 2.3.

TTF is the most liquid natural gas market in Europe, with traded volumes 4.5 times the combined volume of the other eight main European hubs (Heather 2026). Therefore, the TTF gas price serves as the reference price for gas contracts.

At virtual hubs such as TTF, gas price formation is determined by short-run supply and demand (Obadi and Korcek 2020). In the short run the infrastructure of the gas grid is fixed, and the supply of gas cannot expand in response to a change in demand. Therefore, the adjustment takes place through the price, rather than the quantity.

Gas demand originates in three sectors. The first source is residential and commercial heating, which makes consumption seasonal. Deviations of weather from the seasonal norm also affect prices (Brown and Yücel 2008). Industrial production is the second source of demand. In the chemical sector, for example, gas is used as an energy source and as a raw material. The third source is power generation, where gas competes with coal. As a result, the gas market is linked to the coal market and, through emission allowances, to the carbon market.

On the supply side, natural gas prices depend on pipeline flows and deliveries of liquefied natural gas (LNG). LNG is natural gas cooled to liquid form for transport by ship.

As has been observed repeatedly in the past, the European natural gas market is highly sensitive to supply disruptions. The European gas price crisis between 2021 and 2022 is the most severe episode. During the COVID-19 pandemic in 2020, energy

demand collapsed. This was followed by economies reopening, and gas demand recovered faster than production could follow. From mid-2021, the post-pandemic increase in demand was also combined with reduced Russian pipeline flows, strong Asian competition for LNG cargoes and storage levels below the seasonal norm (Heather 2022). As a result, TTF gas prices rose from roughly 18 EUR/MWh in the first quarter of 2021 to a peak of 339 EUR/MWh on 26 August 2022 and then declined from early 2023.

The crisis eventually raised the question of whether the TTF price benchmark had become less representative of the underlying gas market. In November 2022, the European Commission proposed a Market Correction Mechanism (MCM), which would cap gas prices during crisis periods (European Commission 2022). The mechanism applied until January 2025 but was never activated, as gas prices declined after 2023.

A more recent disruption followed the military campaign against Iran by Israel and the United States on 28 February 2026. Iran responded with attacks on shipping and energy infrastructure in the Gulf. Between 27 February and 2 March, the traffic through the Strait of Hormuz fell from 95 vessels to 4 vessels per day (Sharples and Farren-Price 2026). These events resulted in a spike of TTF gas prices from 27 EUR/MWh on 5 January 2026 to 61 EUR/MWh on 19 March 2026.

2.3 Futures Contracts

A derivative is a contract whose value depends on the price of another asset, called the underlying. A forward contract is an agreement between two parties to buy or sell an asset at a fixed price on a specified future date. A futures contract is a standardized forward contract traded on an exchange (Hull 2022, pp. 28–30). The quantity and the delivery date are set by the exchange. The price is agreed between the parties and remains fixed for the contract. As a result, all contracts for a given month are identical in their terms and can be freely transferred. TTF gas futures are quoted in EUR/MWh and are physically delivered, with delivery spread evenly across the hours of the delivery month. Futures contracts are settled daily against a settlement price published by the exchange. The settlement price is determined by the exchange at the end of each trading day and is not necessarily a price at which a transaction occurred. The daily gain or loss is exchanged in cash between the two parties.

The payoff of a long position in the futures contract is the difference between the settlement price F_T at expiry T and the agreed price K . The payoff of a short position is the difference between the agreed price K and the settlement price F_T . Both payoffs can be positive or negative (Hull 2022, p. 29).

Price formation of natural gas in Europe has generally taken place on the futures rather than spot market (Obadi and Korcek 2020). In a spot market the gas is traded for immediate delivery. A futures contract can be traded repeatedly before expiry. As a result, the market is open for hedgers and speculators without an interest in the physical commodity.

At any point in time, contracts for multiple consecutive months are listed and can be traded in parallel. The contract with the nearest delivery on a given day is the front-month contract (M1), and the next is the second-month contract (M2). Trading concentrates in the front-month contract, hence it is commonly used as the reference for the current gas price. Since each contract ceases trading before its delivery month, the contract that is the front month changes over time. The prices of the contracts sorted by the delivery month form the term structure of the gas market. The market is in contango if the term structure is upward sloping, and in backwardation if it is downward sloping. The price difference between two consecutive months is the calendar spread. A calendar spread describes the shape of the term structure.

2.4 Options

An option is a derivative that gives its holder the right, but not the obligation, to buy or sell an underlying asset at a fixed price. Options are traded on exchanges and also directly between two parties, which is referred to as over-the-counter trading. A **call option** is a right to buy the underlying asset and a **put option** is a right to sell it. The fixed price is called the strike price K , and the date on which the right expires is the expiry date.

American options can be exercised at any point up to the expiry date. European options can only be exercised on the expiry date (Hull 2022, p. 31). TTF gas options are European.

The payoff of an option is the value of the right at expiry. For a call option, the payoff is $\max(F_T - K, 0)$, where F_T is the settlement price at expiry. The right is exercised only if $F_T > K$, otherwise the option is left unused. For a put option, the payoff is $\max(K - F_T, 0)$. The payoff of an option is never negative (Hull 2022, p. 231). When the option is bought, the buyer pays the seller an amount called the premium. The premium is the price of an option.

The relation between the current price and K is described by three terms. A call option is **in the money** if $F_t > K$, **at the money** if $F_t = K$ and **out of the money** if $F_t < K$, where F_t is the settlement price on trading day t . For a put option, the definitions are reversed (Hull 2022, p. 234).

The settlement price at expiry is unknown when the option is bought. A risk-neutral world is a world in which investors do not require a higher expected return for taking on more risk. In a risk-neutral world, the expected return on any investment is the risk-free rate. Pricing an option under this assumption gives the correct price in the real world as well (Hull 2022, p. 292). The price of a European call is its expected payoff in a risk-neutral world, discounted at the risk-free rate

$$c = e^{-rT} \hat{\mathbb{E}}[\max(F_T - K, 0)], \quad (2)$$

and the price of a European put is

$$p = e^{-rT} \hat{\mathbb{E}}[\max(K - F_T, 0)] \quad (3)$$

where $\hat{\mathbb{E}}$ denotes the expected value in a risk-neutral world, r is the risk-free rate and T is the time to maturity. The option is valued at time zero, so that the expiry date and the time to maturity are both equal to T . The probabilities entering $\hat{\mathbb{E}}$ are those of a risk-neutral world and not those observed in the price series. The price of an option depends on the distribution of F_T , rather than on its expected value alone.

An option that is in the money would pay a positive amount at the current price, while an option that is out of the money would pay nothing. As a result, the value of an option depends entirely on the probability that the price moves past the strike before expiry.

2.5 Implied Volatility

As described in Sec. 2.4, the price of an option depends on the distribution of F_T in the risk-neutral world. Black (1976) gives a closed-form solution for Eq. (2) and Eq. (3) under the assumption that F_T is lognormal. The result is known as the Black-76 model. The model further assumes that σ and r are constant over the life of the option and that the option is European.

A random variable X is normally distributed with mean μ and standard deviation σ if its probability density is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right). \quad (4)$$

The distribution is symmetric around μ and is fully described by μ and σ . For the standard normal distribution $\mu = 0$ and $\sigma = 1$. A random variable X is lognormally distributed if $\ln X$ is normally distributed.

Under the lognormal assumption, the European call and put prices are given by

$$c = e^{-rT} [F_0 N(d_1) - K N(d_2)], \quad (5)$$

$$p = e^{-rT} [K N(-d_2) - F_0 N(-d_1)], \quad (6)$$

where N is the cumulative distribution function of the standard normal distribution, F_0 is the current futures price, K is the strike price, T is the time to maturity, r is the risk-free rate, and σ is the volatility of the futures price, as defined in Sec. 2.1. d_1 and d_2 are given by

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (7)$$

The parameters entering Eq. (5) and Eq. (6) are observable, except for σ . The implied volatility is the value of σ that makes the model price equal the market price of the

option. Eq. (5) and Eq. (6) cannot be solved for σ in closed form, since σ enters through N and appears in both d_1 and d_2 . As a result, the implied volatility is found numerically. The option price increases with σ , so exactly one value reproduces the observed price (Hull 2022, p. 358).

If the Black-76 model held, the implied volatility of options on the same futures contract with the same expiry would be constant across strikes. The plot of implied volatility against K is known as the volatility smile. In practice the smile is not flat. The distribution of F_T implied by the market prices of options, referred to as the implied distribution, therefore departs from the lognormal (Geman 2006, pp. 101–102).

In equity markets, the implied volatility decreases as the strike price increases, since equity prices and volatility are negatively correlated (Hull 2022, p. 456). In energy markets the smile is skewed to the right, since participants pay more for protection against a sharp rise in prices. This was found for crude oil, heating oil, gasoline and natural gas (Geman 2006, p. 105).

TODO: Somewhere in the section: state that Black-76 is used as a reference price and a change of units, not because TTF prices are believed lognormal. Its departure from the data is what makes the smile informative.

2.6 Stylized Facts of Asset Returns

A diverse collection of assets, markets and time periods share common statistical characteristics of returns. These characteristics are referred to as stylized facts. Eleven stylized facts are summarized in Cont (2001). As described in Sec. 2.4, the value of an option depends on the distribution of F_T . Only three stylized facts describing this distribution are measured for TTF: heavy tails, volatility clustering and asymmetry.

Heavy tails. The tails of a distribution are the regions far from its mean. Large and infrequent returns are located in the tails. The distribution of financial returns has more mass in the tails than a normal distribution (Cont 2001). This is referred to as heavy tails. The weight of the tails is measured by the excess kurtosis

$$\text{Kurt}_{\text{ex}}(X) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] - 3, \quad (8)$$

where μ and σ are the mean and the standard deviation of the random variable X . The excess kurtosis of a normal distribution is 0. The distribution is considered heavy-tailed if its excess kurtosis is positive. The tails of the return distribution determine the value of out-of-the-money options, which pay nothing unless a large move occurs. A volatility smile with higher implied volatility at strikes far from the current price, in both directions, corresponds to an implied distribution (Sec. 2.5) with heavier tails than the lognormal (Hull 2022, p. 453).

Volatility clustering. Large returns are more likely to be followed by large returns, and small returns by small returns (Cont 2001). This is measured with the autocorre-

lation. Autocorrelation is the correlation of a time series with itself, shifted by a fixed number of observations called the lag. The autocorrelation of returns at lag τ is

$$C(\tau) = \text{corr}(x_{t+\tau}, x_t), \quad (9)$$

where x_t is the return on trading day t . The returns themselves show almost no autocorrelation, while absolute or squared returns are positively correlated (Cont 2001). Volatility clustering is measured on squared returns

$$C_2(\tau) = \text{corr}(x_{t+\tau}^2, x_t^2). \quad (10)$$

C_2 remains positive through squaring and decays slowly over several days. The distribution of F_T covers the life of the option rather than over a single day. As the length of the period over which the returns are calculated increases, the distribution of returns approaches a normal distribution. This property is referred to as aggregational Gaussianity (Cont 2001). Volatility clustering slows the convergence to a normal distribution.

Gain/loss asymmetry. The distribution of returns is asymmetric around its mean. Asymmetry is measured with the skewness

$$\text{Skew}(X) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right], \quad (11)$$

where μ and σ are the mean and the standard deviation of the random variable X . The skewness of a normal distribution is 0. The skewness is positive when large positive returns are more likely than large negative returns, and negative in the opposite case. The distribution is considered asymmetric if its skewness is non-zero. A volatility smile that is higher on one side than the other corresponds to an implied distribution with a heavier tail on that side (Hull 2022, p. 456). The skewness of the implied distribution is therefore non-zero.

Eq. (11) and Eq. (8) are defined over a random variable X . In practice, the distribution of returns is unknown and only a sample of n observations $x_1, \dots, x_n$ is available. Therefore, the mean μ and the standard deviation σ are computed from the sample, and $\mathbb{E}[\cdot]$ becomes an average over the n observations. The resulting sample estimators of skewness and excess kurtosis are

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{3/2}} \quad \text{and} \quad g_2 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{\left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^2} - 3, \quad (12)$$

where $\bar{x}$ is the mean of the n observations.

Mean reversion and seasonality are commodity-specific stylized facts (Geman 2006, pp. 64–68). Mean reversion is the tendency of a price to return to its long-term average after a period of deviation. Seasonality is the tendency of a price to follow a predictable pattern over the course of a year. These stylized facts describe the behavior of the price level over months and years, while this thesis models the distribution of daily returns.

3 Machine Learning Fundamentals

This section provides the machine learning background for the remainder of the thesis. It begins with the terminology of neural networks and their training. The section then focuses on recurrent architectures and the long short-term memory (LSTM) cell. The generative adversarial network (GAN) is introduced next, followed by its conditional, forecasting and financial variants.

3.1 Neural Networks

3.1.1 Recurrent Neural Networks and LSTM

3.2 Generative Adversarial Networks

3.2.1 Basic Generative Adversarial Networks

Generative modeling is an unsupervised learning approach. A generative model takes a set of training examples and learns the probability distribution that produced them. GANs are generative models first introduced by I. J. Goodfellow et al. (2014). GANs have been particularly successful in image and video generation. They have also been used to create synthetic data, for training other machine learning models and to fill in gaps in missing data (I. Goodfellow et al. 2020).

GANs are based on game theory, which studies situations where several decision-makers are present. Each participant chooses an action, and the outcome depends on the choices of all of the participants. A GAN has two decision-makers, **the generator (G)** and **the discriminator (D)**, typically implemented as neural networks. G takes a vector of random noise as input and generates a sample as output. D takes a real or generated data sample as input and outputs a probability that the input sample is real. The goal of G is to fool D by generating realistic-looking data samples. The goal of D is to distinguish real samples from generated ones.

Assume that the data x follows an unknown distribution p_{data} , from which new samples need to be generated. The generator is a neural network $G(z; \theta_g)$, where θ_g denotes the learnable parameters. The input z is a noise vector drawn from a prior distribution p_z . Gaussian distribution is normally used to define p_z ; the GAN framework does not add limitations. The probability distribution of the generated samples is denoted p_g . The discriminator is a neural network $D(x; \theta_d)$, with learnable parameters θ_d . It outputs the estimated probability that the input was drawn from p_{data} rather than p_g . G and D have cost functions, respectively $J^{(G)}(\theta_d, \theta_g)$ and $J^{(D)}(\theta_d, \theta_g)$. Both G and D need to minimize their cost functions, while they can only control their learnable parameters, respectively θ_g and θ_d . Hence, the solution is a Nash equilibrium, rather than a minimum of a single cost function.

The GAN framework is flexible in the sense of the cost functions applied to G and D . The simplest version of the game is a zero-sum game. In this, the sum of all of the cost functions is always zero.

$$J^{(G)}(\theta_d, \theta_g) = -J^{(D)}(\theta_d, \theta_g). \quad (13)$$

The cost function of D is defined as a standard cross-entropy

$$J^{(D)}(\theta_d, \theta_g) = -\frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \log D(x) - \frac{1}{2} \mathbb{E}_z \log(1 - D(G(z))). \quad (14)$$

Learning the zero-sum game minimizes the Jensen-Shannon divergence between p_{data} and p_g . Convergence holds only if both players are updated in function space. In function space D and G are updated freely among all possible functions. In practice, D and G are neural networks and updated in parameter space. In parameter space the reachable functions are limited to those the network architecture can represent. D minimizes a cross-entropy while G maximizes the same quantity. Generated samples are rejected with high confidence at the start of training, and the gradient of G vanishes at the point at which it performs worst (I. Goodfellow 2017).

Therefore, the cost function of G is constructed by flipping the target of the cross-entropy.

$$J^{(G)}(\theta_d, \theta_g) = -\frac{1}{2} \mathbb{E}_{p_z} [\log[D(G(z; \theta_g); \theta_d)]] . \quad (15)$$

The cost function of D and G do not sum to zero anymore, hence the game is no longer zero-sum. Alternative cost functions have been introduced over time.

Fig. 1 is the visualization of a standard GAN workflow adapted from Vuletić et al. (2024).

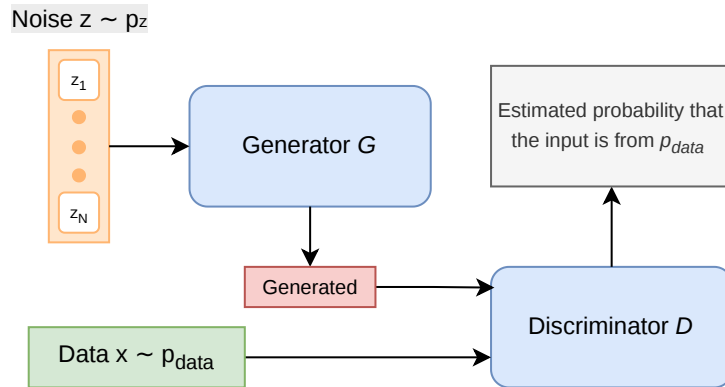


Figure 1: Structure of a standard GAN

GANs are trained by updating the two networks in turn. At each step, a minibatch of samples and noise vectors are drawn from p_{data} and p_z respectively. D is updated

with respect to θ_d while θ_g is fixed. G is updated with respect to θ_g while θ_d is fixed. D is commonly trained k times for every update of G , although I. Goodfellow (2017) argues that $k = 1$ is sufficient. k is defined as an independent training parameter.

As stated previously, convergence is not guaranteed in practice. This means that the training of a GAN can settle in a state where $p_g \neq p_{\text{data}}$. The most common such state is mode collapse. It occurs when G repeatedly maps noise vector to very similar outputs. As a result, p_g covers only a small part of p_{data} but D assigns a high probability that the generated samples are real. Mode collapse is still an unresolved challenge in GAN training. In recent years, several strategies have been proposed to mitigate its effect, such as stronger generators and discriminators, longer training and regularization. However, each of these approaches comes with drawbacks (Barsha and Eberle 2025).

3.2.2 cGANs

Standard GANs can be extended to conditional models (I. J. Goodfellow et al. 2014). This extension was further developed by Mirza and Osindero (2014). In Conditional Generative Adversarial Networks (cGANs), the data follow a conditional distribution $p_{\text{data}}(x \mid c)$, where c is auxiliary deterministic information. The condition is applied to D and G as an additional input. The cost functions for G and D are the same as for a standard GAN with the condition c .

$$J^{(D)}(\theta_d, \theta_g) = -\frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \log D(x \mid c) - \frac{1}{2} \mathbb{E}_z \log(1 - D(G(z \mid c) \mid c)). \quad (16)$$

Fig. 2 is the visualization of a cGAN workflow adapted from Vuletić et al. (2024).

3.2.3 ForGAN

The conditional input proposed in cGANs by Mirza and Osindero (2014) prepared the ground for domain-specific GAN architectures. ForGAN was introduced specifically for time series forecasting (Koochali et al. 2019).

ForGAN is trained to approximate the conditional distribution of the next value in a time series. The condition is the sequence of the previous l observations, ending at time t

$$c = (x_{t-(l-1)}, \dots, x_t). \quad (17)$$

A sliding window of $l + 1$ observations is moved along the series. At each position, the first l observations form c and the last is x_{t+1} . G takes a noise vector z and c and outputs a value for the target x_{t+1} . D takes a value of x_{t+1} and c and outputs the probability that x_{t+1} is from $p(x_{t+1} \mid c)$.

A full probabilistic distribution of x_{t+1} can be obtained through repeated sampling of G with different noise vectors and the same c .

ForGAN adds a recurrent layer to G and D . The layer processes c as an ordered sequence and constructs a vector representation. The recurrent layer can be a long

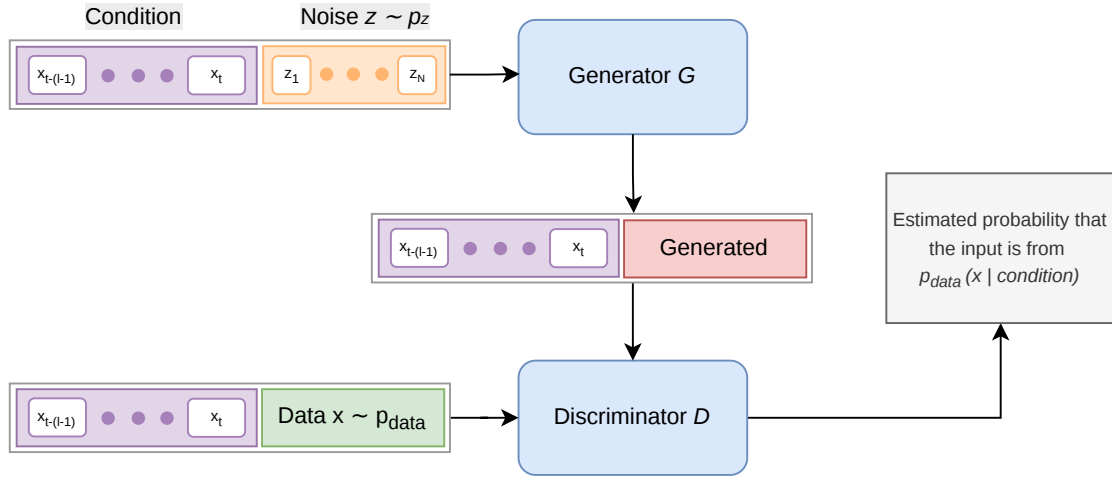


Figure 2: Structure of a cGAN

short-term memory (LSTM)(Hochreiter and Schmidhuber 1997) or a gated recurrent unit (GRU) (Cho et al. 2014). The high-level architecture of the ForGAN model is shown in Fig. 3.

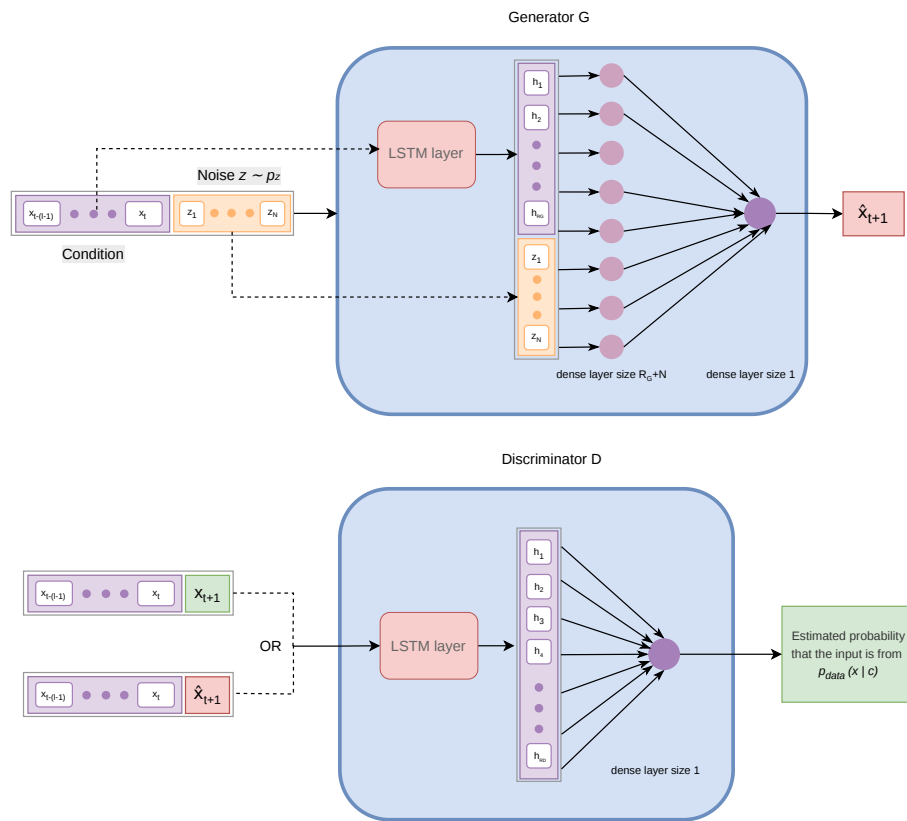


Figure 3: Structure of a ForGAN

3.2.4 Fin-GAN

Fin-GAN was introduced as an extension of ForGAN (Vuletić et al. 2024). It focuses on the probabilistic forecasting of financial time series, specifically equity returns. The ForGAN architecture and the cost function of D are unchanged. LSTM is chosen as the recurrent layer.

The motivation behind Fin-GAN is to correctly forecast the sign of the return, since the sign determines whether the trade is profitable. For this purpose, economics-driven terms are introduced into the cost function of G . In contrast to the unsupervised setting of standard GANs, Vuletić et al. (2024) classify Fin-GAN as supervised learning.

The terms are defined over a minibatch of n_{batch} pairs. For each condition in the batch, G produces a forecast $\hat{x}_i$ of the realized value x_i . Let $\hat{x} = (\hat{x}_1, \dots, \hat{x}_{n_{\text{batch}}})$ and $x = (x_1, \dots, x_{n_{\text{batch}}})$.

Cost function. $J^{(G)}$ can be the zero-sum, cross-entropy or Wasserstein cost function. The latter replaces the Jensen–Shannon divergence between p_{data} and p_g with the Wasserstein-1 distance (Arjovsky et al. 2017). Vuletić et al. (2024) report that the Wasserstein cost function was numerically unstable during training. Hence, $J^{(G)}$ is the cross-entropy cost function given in Eq. (15). In addition, Fin-GAN introduces four economics-driven terms to the cost function. These are defined below.

$$L^G(x, \hat{x}) = J^{(G)}(x, \hat{x}) - \alpha \text{PnL}_a^*(x, \hat{x}) + \beta \text{MSE}(x, \hat{x}) - \gamma \text{SR}^*(x, \hat{x}) + \delta \text{STD}(x, \hat{x}). \quad (18)$$

A positive forecast corresponds to buying an asset, and a negative forecast to selling it short. The profit and loss PnL of the trade is the sign of the forecast multiplied by the realized return $\text{PnL}(x_i, \hat{x}_i) = \text{sgn}(\hat{x}_i) \cdot x_i$. Since only the sign enters the formula, the outcome does not depend on the magnitude of the forecast. $\text{sgn}(\hat{x}_i)$ is not differentiable, so the standard term PnL would not provide a gradient for training. A smooth approximation is introduced

$$\text{PnL}_a^*(x_i, \hat{x}_i) = \tanh(k_{\text{tanh}} \hat{x}_i) x_i, \quad (19)$$

where k_{tanh} is a hyperparameter that controls the accuracy of the approximation. Vuletić et al. (2024) use $k_{\text{tanh}} = 100$, that is large enough to give a close approximation and small enough to maintain strong gradients.

Although the sign of the forecast is the main interest, forecasting an $\hat{x}_i$ that is close to x_i is also of importance. The mean squared error MSE measures the squared difference between the forecast and the realized value, averaged over the minibatch.

SR^* is defined as the mean of the PnL_a^* series divided by its standard deviation.

$$\text{SR}^*(x, \hat{x}) = \frac{\bar{p}}{\sqrt{\frac{1}{n_{\text{batch}}} \sum_{i=1}^{n_{\text{batch}}} (p_i - \bar{p})^2}}, \quad (20)$$

where $p_i = \text{PnL}_a^*(x_i, \hat{x}_i)$ and $\bar{p}$ is its mean over the n_{batch} forecasts. SR^* deviates from Eq. 1. The risk-free rate is not subtracted, since the daily risk-free rate is small

relative to the daily PnL_a^* . The ratio is not annualized, because a constant factor would be absorbed by the weight γ during gradient matching. Lastly, SR^* is computed over the n_{batch} forecasts of a minibatch rather than over a return series. Stochastic gradient descent requires random minibatches. As a result, forecasts entering SR^* are not in chronological order (Vuletić et al. 2024, p. 11).

The standard deviation STD term is the denominator of Eq. 20. Logically maximizing SR^* is the same as maximizing PnL_a^* and minimizing STD at the same time. As a result, including all three terms in L^G is redundant. However, during training SR^* and the combination of PnL_a^* and STD produce different gradients (Vuletić et al. 2024).

Gradient matching. All the terms above are measured in different units, hence they produce incomparable gradients. As a solution, four additional hyperparameters $\alpha, \beta, \gamma, \delta$ are introduced. Fin-GAN is first trained on $J^{(G)}$ only, and gradient norms of other terms are calculated. Each hyperparameter is the mean of the ratios between the cross-entropy gradient norms and the gradient norms of the corresponding term. Large values of $\alpha, \beta, \gamma, \delta$ make the feedback of D insignificant and move the model towards supervised learning, but small values leave it in the standard GAN setting.

If the term is not included in the objective, the hyperparameter is set to zero. Several versions of L^G can be constructed from different combinations of terms. Only combinations that include economics-driven terms are considered for Fin-GAN. Vuletić et al. (2024) train eight such versions and select the combination with the highest Sharpe ratio SR_w on the validation set. Then, the selected G is evaluated on the test set.

Mode collapse. Vuletić et al. (2024) distinguish two types of mode collapse for time series. Distributional mode collapse happens when G generates nearly identical samples for a single condition. Mean mode collapse happens when the means of the generated distributions are nearly identical across all conditions. Both are identified by a standard deviation below a threshold $\epsilon = 0.0002$. Fin-GAN uses normal Xavier initialization (Glorot and Bengio 2010), since ForGAN suffered mode collapse in 67% of runs under He initialization (He et al. 2015). However, no mode collapse happened with the economics-driven terms, under either initialization.

Weighted strategy. The Sharpe ratio SR_w is not to be confused with the SR^* term of Eq. 20. SR_w is computed over a full data split on wPnL. SR_w measures the directional accuracy of the forecasts and not the shape of the generated distribution. For each condition, B samples are drawn from G with different noise vectors.

$$\text{wPnL}_2^i = 10000 (p_u^i - p_d^i) x_i, \quad (21)$$

where p_u^i and p_d^i are the fractions of non-negative and negative samples. Vuletić et al. (2024) use $B = 1000$. Since the equity data uses two returns per day, intraday and overnight, two positions are taken per day. The wPnL^i series is the sum of the two

wPnL_2^i values per day. For the equity data of Vuletić et al. (2024), two returns per day give $n_{\text{days}} = \lfloor n/2 \rfloor$. The Sharpe ratio SR_w is then

$$\text{SR}_w = \sqrt{252} \frac{\overline{\text{wPnL}}}{\sqrt{\frac{1}{n_{\text{days}}} \sum_{i=1}^{n_{\text{days}}} (\text{wPnL}^i - \overline{\text{wPnL}})^2}}, \quad (22)$$

where $\overline{\text{wPnL}}$ is the mean of the wPnL^i series over the n_{days} trading days. The factor $\sqrt{252}$ annualizes the ratio, where 252 is the conventional number of trading days in a year.

3.3 Related Work

TODO: might need Seasonality and spikes in the natural gas market Francesco Rotondi

4 Methodology and Implementation

This section describes the methodology and implementation. It begins with the construction of the TTF return series from the source data and the preparation of the options data, followed by the description of the model architecture, the training setup and the adaptations to the reference Fin-GAN implementation. Next, the experimental design and benchmark models are introduced. The evaluation method for the generative model is described, followed by the setup for the option pricing procedure. The section closes with the software environment.

4.1 Data Pipeline

Two datasets are used in this thesis. The first is the TTF futures dataset, which is used to train and evaluate the model. The second is the TTF options dataset, which provides the market prices for the benchmark in Sec. 4.6.

4.1.1 Futures Data

The futures data was obtained from BASF's Bloomberg infrastructure and covers TTF natural gas futures for the first three nearby contracts between January 2010 and April 2026. The data is distributed as three files: contract prices, contract metadata, and monthly listings of active contracts.

Daily settlement prices. This file provides the price history of the contracts in the dataset. Each contract is observed once per trading day, identified by contract symbol and trading date. Each record contains opening, high, low and daily closing prices in EUR/MWh, trading volume and open interest. Coverage of these fields varies considerably with how actively a contract was traded and how far from expiry it is. The closing price is the exception and is available throughout the whole dataset. Additionally, it falls outside the intraday high–low range for 7,712 of 25,884 records, or 29.8%. The deviation is small in most cases, with a median of 0.10 EUR/MWh, but reaches up to 42.26 EUR/MWh. The latter confirms that it is a settlement price set by the exchange rather than a price at which trading occurred.

Contract metadata. This file provides static attributes per unique contract in the dataset. The expiry date provided per record is used to rank contracts by time to maturity in Sec. 4.1.2. All contracts in the dataset are physically delivered instruments traded on ICE Endex and quoted in EUR/MWh. First and last trading dates, notice dates and delivery dates are also recorded.

Contract listings. This file records which contracts were active on the market, as monthly snapshots taken on the first calendar day of each month. Each contract is observed once per snapshot. Together with the expiry dates, the listings determine which contract is the front month on a given date, as described in Sec. 4.1.2.

4.1.2 Front-Month Series Construction

As described in Sec. 4.1.1, the contract listings and the contract metadata each hold part of the information required to identify the front-month contract. The listings record which contracts were active in a given month, while the metadata records when each contract expires. Therefore, the expiry dates were joined onto the listings by contract symbol. As a result, each listing record carries the expiry date of the contract it refers to.

The contracts were then sorted by expiry date within each monthly snapshot and the earliest-expiring one was kept.¹ This gives the front-month contract for that month. Since the snapshots are monthly but the series must be daily, each trading day was assigned the front-month contract from the most recent snapshot on or before that day. The settlement price of the selected contract was then added for each trading day, matching on trading date and contract symbol. The trading days without an available settlement price were discarded. The resulting series contains 3,998 trading days spanning 2010-01-04 to 2026-04-24, with settlement prices ranging from 3.509 to 339.196 EUR/MWh.

Since the front-month contract is replaced at each expiry, the series is not the price history of a single instrument. Returns spanning a contract transition reflect the change in the price level as well as the difference between the two contracts. Across the 195 transition days in the sample, the standard deviation of daily log returns is 0.071, compared to 0.037 on all other days. When the transition days are excluded, excess kurtosis falls from 14.91 to 14.65, and skewness from 0.70 to 0.26.² Transition days contribute little to the tail weight of the series, but they are responsible for most of its positive skew.

The transition effect described above is preserved and quantified rather than removed, since restricting the analysis to a single contract would reduce the sample to a few months.

4.1.3 Return Definition

The original Fin-GAN input series is built from two returns per day (Vuletić et al. 2024). The first is the intraday return, from the opening to the closing price and the second is the overnight return, from one closing price to the next opening price. The return of the corresponding sector ETF is then subtracted from each entry. This isolates the part of the movement specific to the asset.

Neither step has been adopted for the TTF futures dataset. The two-return scheme requires opening and closing prices that are both traded prices. As shown in Sec. 4.1.1, the daily price in this dataset is a settlement price set by the exchange. A return computed between the opening trade and the settlement price would therefore combine two different price concepts.

¹Trading in a TTF futures contract ceases two UK business days before the first day of the delivery month.

²Skewness and excess kurtosis are computed with the sample estimators g_1 and g_2 of Eq. (12).

Additionally, the excess-return step requires a benchmark that leaves a meaningful residual when subtracted. In the case of equities, subtracting the sector ETF return removes the market-wide component and leaves the movement specific to the individual stock. European natural gas has no comparable instrument.

As a substitute, the second nearby contract was considered. However, the difference between the first and second nearby returns is a calendar spread return, as described in Sec. 2.3. A model that is trained on it would learn how the gap between two contracts moves, not how the gas price itself moves. Option payoffs depend on the price level at expiry, so a model trained on spread returns could not be used to price options.

Therefore, the dataset was constructed from daily logarithmic returns of the front-month settlement price. Returns are approximately stationary and therefore comparable across the sample period. Additionally, log returns are additive across time and allow price levels to be recovered exactly from simulated returns. The latter is required for the Monte Carlo procedure in Sec. 4.6.

$$x_t = \log\left(\frac{P_t}{P_{t-1}}\right), \quad (23)$$

where P_t is the front-month settlement price on trading day t . The transformation gives 3,997 return observations.

4.1.4 Data Split

The return series was split chronologically into training (80%), validation (10%), and test (10%) sets using the proportions of Vuletić et al. (2024). Training covers 2010-01-05 to 2023-01-18 validation 2023-01-19 to 2024-09-04 (399 returns), and test 2024-09-05 to 2026-04-24 (401 returns). The chronological split was explicitly used to avoid look-ahead bias. Following the current split, the 2021 to 2022 price disruption discussed in Sec. 2.2 falls entirely within the training period, and the 2026 disruption falls entirely within the test period, as seen in Fig. 4.

Sliding window matrices were constructed for each split separately, so that no window spans a split boundary. The window was moved forward by one trading day.

4.1.5 Options Data

4.2 Architecture, Setup and Adaptations

4.2.1 Model Architecture

The ForGAN architecture described in Sec. 3.2.3 was used without structural modifications. In G , the condition passed through an LSTM layer. The final hidden state was concatenated with the noise vector z . The result passed through a dense layer of width $R_G + N$, then a ReLU activation and another dense layer of width one, where R_G is the LSTM hidden size and N is the noise dimension. In D , the candidate value ($\hat{x}_{t+1}$ or x_{t+1})

4 Methodology and Implementation

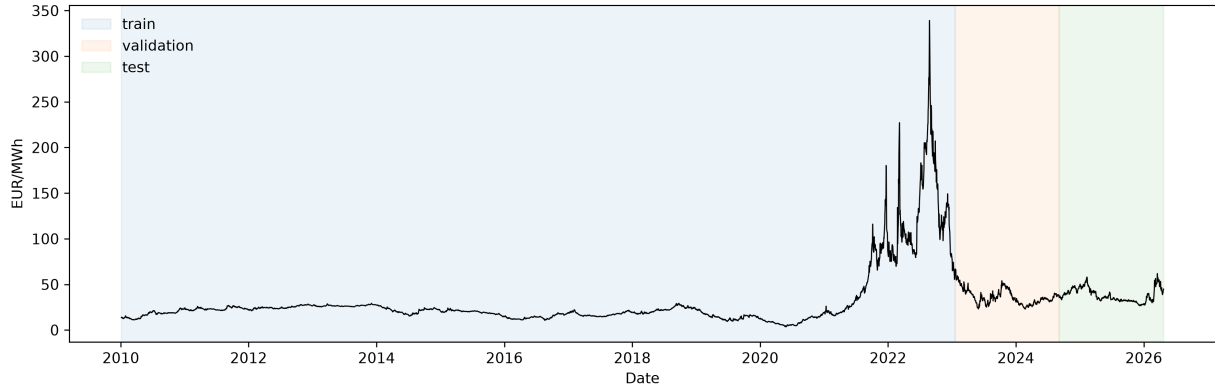


Figure 4: TTF train, validation and test data split

was appended to the condition, which gave a sequence of size $l + 1$. The result passed through an LSTM layer with the hidden size R_D . The final hidden state passed through a dense layer with a sigmoid activation.

Following Vuletić et al. (2024), $N = 8$, $R_G = 8$, $R_D = 8$ and $l = 10$. One value was forecast per condition, $\hat{x}_{t+1}$. However, the parameters N , R_G , R_D , and l were varied in Sec. 4.3. The normal Xavier initialization described in Sec. 3.2.4 was used.

4.2.2 Training configuration

The RMSProp optimizer was used for both G and D , with a learning rate of 0.0001. One discriminator update was performed for each generator update, as described in Sec. 3.2.1. The weights were updated after each minibatch of size $n_{\text{batch}} = 100$. The training data was shuffled at the start of each epoch.

The input to G and D was standardized using the mean and standard deviation of the first 100 rows of the training window matrix. Using the earliest rows avoids look-ahead bias. As described in Sec. 4.1.4, the training split starts in January 2010. This means that the reference batch covers a period of relatively low volatility. The scaling parameters are not representative of the whole dataset, especially the 2021 to 2022 period of high volatility. The effect of this choice was tested separately, as described in Sec. 4.3.

The training was performed in two phases. In the first phase, both G and D were trained on cross-entropy alone for n_{grad} epochs. The gradient norms were recorded and α , β , γ and δ were computed as described in Sec. 3.2.4. In the second phase, the networks were trained for n_{epochs} further epochs, separately for each of the nine cost function combinations listed in Sec. 4.3. An additional tenth branch was trained on cross-entropy alone for n_{epochs} further epochs, giving the ForGAN baseline used for comparison in Chapter 5. The effective training length was $n_{\text{grad}} + n_{\text{epochs}}$, since the network weights were updated in both phases. The values n_{grad} and n_{epochs} were varied across configurations, as described in Sec. 4.3.

Following Vuletić et al. (2024), the cost function combination and the corresponding G with the highest SR_w on the validation data split were chosen. All combinations were further evaluated on the test split. SR_w measures directional accuracy, while option prices depend on the shape of the return distribution. Therefore, the generated returns were additionally evaluated on distributional statistics in Sec. 4.5.

4.2.3 Adaptations

The implementation is based on the reference Fin-GAN code published by Vuletić et al. (2024). The network definitions, the gradient norm matching procedure and the training loops were adopted from that implementation. The cost function terms of Sec. 3.2.4 were adopted without modification.

The data structure with intraday and overnight returns was not adopted. As described in Sec. 4.1, the dataset was constructed from daily log returns of the front-month settlement prices of TTF futures. The data loading functions were modified accordingly. Additionally, a separate evaluation module was implemented to compute SR_w , the forecast accuracy and the distributional statistics, as described in Sec. 4.5. A new SR_w function was required, because the original implementation sums two $wPnL_2^i$ values per day, as described in Sec. 3.2.4. This is not applicable to the daily TTF data.

G produces one value per noise sample, $\hat{x}_{t+1}$. To generate price paths, recursive rollout was implemented, as described in Sec. 4.6.

4.3 Experimental Design

As described in Sec. 3.2.4, the eight cost function combinations of Vuletić et al. (2024) were used: PnL_a^* ; SR^* ; PnL_a^* & MSE; PnL_a^* & SR^* ; PnL_a^* & STD; SR^* & MSE; PnL_a^* & MSE & SR^* ; PnL_a^* & MSE & STD.

A ninth combination, using only the MSE, was added as a supervised counterpart to ForGAN. A tenth branch was trained on cross-entropy alone, giving the ForGAN baseline. Each combination was trained under several configurations, listed in Table 1.

The configurations differ in one parameter at a time, to isolate the effect of each change. The first configuration is the reference configuration, following the values and the schedule of Algorithm 1 in Vuletić et al. (2024). The second configuration tests whether a longer training period improves the distributional quality. The third and fourth configurations increase N and R_D separately. The fifth configuration increases both, matching the default values of the published implementation. The sixth configuration tests whether a longer conditional window of 30 trading days carries more information about the volatility of the series. The seventh configuration extends the training further. The eighth configuration increases R_G . R_G is not varied in the paper or in the published implementation. The ninth configuration combines the longer training with the increased R_G . Lastly, the tenth configuration combines the increased R_G with the original N . An additional test run was performed with the scaling parameters computed from the entire training split, using configuration 5 of Table 1.

Configuration	l	N	R_G	R_D	$n_{\text{grad}} + n_{\text{epochs}}$
1	10	8	8	8	25 + 100
2	10	8	8	8	100 + 500
3	10	32	8	8	100 + 500
4	10	8	8	64	100 + 500
5	10	32	8	64	100 + 500
6	30	32	8	64	100 + 500
7	10	32	8	64	100 + 1500
8	10	32	64	64	100 + 500
9	10	32	64	64	100 + 1500
10	10	8	64	64	100 + 500

Table 1: Model configurations

Each configuration was trained with 5 fixed random seeds. Training runs of the same configuration differ noticeably in their distributional statistics, due to the stochastic nature of the training process. Therefore, all results were reported as means and standard deviations across seeds. Fixing the seeds makes the runs reproducible in the environment described in Sec. 4.7.

4.4 Benchmark Models

4.5 Evaluation Methodology

G was evaluated on the test and validation splits. Each row of the window matrix consists of a condition window of l realized returns and the realized return of the next day. The conditions were passed to the trained G and the generated values were compared to the realized values. For each condition, $B = 1000$ samples were drawn from G with different noise vectors.

The realized returns of the split, $x = (x_1, \dots, x_n)$, give the reference values, where n is the number of rows in the window matrix. The generated draws form a matrix $\hat{X} \in \mathbb{R}^{n \times B}$, where $\hat{x}_{i,j}$ is the j -th draw of condition i . Additional sets were derived from $\hat{X}$.

The path set $\hat{x}_{\text{path}} = (\hat{x}_{1,1}, \dots, \hat{x}_{n,1})$ is the first column of $\hat{X}$, giving one generated return per trading day. The pooled set takes all nB generated entries of $\hat{X}$.

$$\hat{x}_{\text{pool}} = \{ \hat{x}_{i,j} \mid i = 1, \dots, n, j = 1, \dots, B \}. \quad (24)$$

The mean set holds the mean of all draws for each condition

$$m = (m_1, \dots, m_n), \text{ where } m_i = \frac{1}{B} \sum_{j=1}^B \hat{x}_{i,j}. \quad (25)$$

$\hat{x}_{\text{path}}$ is directly comparable with x , since both sets have the same length. $\hat{x}_{\text{pool}}$ contains B times more observations than x .

Distributional statistics were computed for x , $\hat{x}_{\text{path}}$ and $\hat{x}_{\text{pool}}$: mean, standard deviation, skewness, excess kurtosis, minimum, maximum, 1% and 99% quantiles. Additionally, the fractions of returns exceeding 5% and 10% in absolute value were computed.

The point forecast accuracy was measured using x and m . The mean absolute error (MAE), the root mean squared error (RMSE) and correlation were computed, following Vuletić et al. (2024). The fraction of positive values in m was calculated to check whether the forecasts are biased in one direction. The standard deviation of m was calculated to check how much the forecasts vary across conditions.

$C_2(\tau)$ from Eq. 10 was computed for x and $\hat{x}_{\text{path}}$ at $\tau = 1, 2, 5, 10$. $\hat{x}_{\text{pool}}$ cannot be used since it has no order in time.

The weighted strategy described in Sec. 3.2.4 was applied to the rows of $\hat{X}$ and x . $\overline{\text{wPnL}}$ and SR_w were computed. Since the TTF series contains one return per trading day, $n_{\text{days}} = n$ in Eq. 22.

Lastly, the mode collapse described in Sec. 3.2.4 was checked. For distributional mode collapse, the standard deviation of each row of $\hat{X}$ was computed. The smallest value was compared with $\epsilon = 0.0002$. For mean mode collapse, the standard deviation of m was compared with the same ϵ value.

4.6 Option Pricing Setup

4.7 Software Environment

Experiments were run on a MacBook Pro with an Apple M4 Pro processor and 24 GB of RAM, running macOS 26.6.1. The NVIDIA CUDA toolkit is not supported on macOS. Training was on CPU. The software packages and their respective versions used for the data pipeline, model training and evaluation are listed in Table 2.

Package	Version
Python	3.12.13
PyTorch	2.13.0
NumPy	2.5.2
pandas	3.0.5
polars	1.43.2
pyarrow	25.0.1
DuckDB	1.5.5
SciPy	1.18.0
Matplotlib	3.11.1

Table 2: Software packages and versions

4 Methodology and Implementation

A fixed seed was set for the Python, NumPy and PyTorch random number generators at the start of each run.

The TTF futures data from Bloomberg and the options data from Databento are licensed and cannot be distributed. The processing steps are documented in Sec. 4.1. The pipeline is reproducible, given access to the same sources.

The implementation is based on the reference code of Vuletić et al. (2024) and is publicly available on GitHub.³

³<https://github.com/milenavuletic/Fin-GAN>

5 Experiments

5.1 Properties of the TTF Return Series

Table 3 reports the distributional properties of the TTF return series, full sample and per data split. The statistics are the reference values against which generated returns are compared in the following sections.

	Full	Train	Validation	Test
n	3,997	3,197	399	401
Mean	0.0003	0.0005	-0.0014	0.0006
Standard deviation	0.0395	0.0377	0.0515	0.0401
Skewness	0.70	0.71	0.59	0.91
Excess kurtosis	14.91	19.08	2.53	13.24
Minimum	-0.3524	-0.3524	-0.1615	-0.1749
Maximum	0.4128	0.4128	0.2483	0.3229
1 % quantile	-0.1153	-0.1129	-0.1248	-0.1291
99 % quantile	0.1239	0.1224	0.1639	0.0885
$P(x_t > 0.05)$	0.117	0.099	0.253	0.130
$P(x_t > 0.10)$	0.030	0.027	0.058	0.025
$C(1)$	0.033	0.067	-0.096	0.008
$C_2(1)$	0.364	0.392	0.128	0.288
$C_2(2)$	0.280	0.310	0.206	0.061
$C_2(5)$	0.215	0.236	0.105	0.094
$C_2(10)$	0.112	0.117	0.148	0.011

Table 3: Distributional statistics and autocorrelation of daily TTF log returns, full sample and per data split

Heavy tails. The excess kurtosis of the full series is 14.91, compared to 0 for a normal distribution (Sec. 2.6). Absolute returns exceed 10 % on 3 % of trading days. The heavy tails are not distributed evenly across the data splits. The training split, which contains the 2021 to 2022 crisis, has an excess kurtosis of 19.08. The test split, which contains the 2026 disruption, has 13.24. Both episodes are described in Sec. 2.2. The validation split does not contain a comparable event and has an excess kurtosis of 2.53.

Asymmetry. The skewness of the full series is 0.70, indicating that large positive returns are more likely than large negative returns (Sec. 2.6). The skewness is positive for all data splits, with the training split at 0.71, the validation split at 0.59 and the test split at 0.91. As reported in Sec. 4.1.2, the transition days account for most of the positive skew. Skewness falls to 0.26 when transition days are excluded. The remaining asymmetry is consistent with the supply disruptions described in Sec. 2.2. Such events

raise prices sharply, while no comparable mechanism lowers them. The positive skew is consistent with the right-skewed volatility smile reported for energy markets in Sec. 2.5.

Volatility clustering. The autocorrelation of returns at $\tau = 1$, Eq. (9), is 0.033 and stays close to 0 across all data splits. The autocorrelation of squared returns, Eq. (10), is 0.364 at $\tau = 1$, and decays slowly, to 0.280 at $\tau = 2$, 0.215 at $\tau = 5$ and 0.112 at $\tau = 10$. The slow decay is the volatility clustering described in Sec. 2.6. The autocorrelation of squared returns is strongest in the training split, which contains the 2021 to 2022 crisis, and weakest in the validation split, which does not contain a comparable event.

Table 4 reports the distributional properties of the full test split and the test split excluding the period from 2026-02-01 to 2026-04-30. The excluded period covers 57 of the 401 trading days. The excess kurtosis of the remaining days is 1.57, the skewness is -0.06 and the autocorrelation of squared returns at $\tau = 1$ is 0.042. Absolute returns exceed 10 % on 0.6 % of trading days, compared to 2.5 % for the full test split.

	Test	Test excluding 2026 disruption
n	401	344
Standard deviation	0.0401	0.0288
Skewness	0.91	-0.06
Excess kurtosis	13.24	1.57
$P(x_t > 0.10)$	0.025	0.006
$C_2(1)$	0.288	0.042

Table 4: Test split statistics with and without the 2026 disruption

When the 2026 disruption is excluded, the test split does not show heavy tails, asymmetry or volatility clustering. The validation split has the highest standard deviation of all splits, and contains no comparable single event. The returns of the validation split are uniformly large, while the returns of the test split are moderate outside a single episode.

5.2 Baseline and Training Length

5.3 Network Capacity

5.4 Condition Window and Input Scaling

5.5 Effect of the Cost Function Terms

5.6 Model Selection

5.7 Option Prices

TODO: Figure: implied volatility against strike, one valuation date, one expiry, 5–7 strikes around the money. Three curves: market (Databento settlement prices), Black-76 with historical volatility from the training split, and Monte Carlo on generated paths. Convert all three prices to implied volatility by inverting Eq. (5), as described in Sec. 2.5. Expected result: the market curve is skewed to the right (Sec. 2.5), Black-76 is flat by construction, and the generated curve is too flat on the call side, since the SR branches thin the upper tail (Sec. 5.1). Prose: state that Black-76 is used here as a change of units, not as a model of the return distribution — prices at different strikes are not comparable in EUR/MWh. Drop-dead date for the Databento leg: September 4. If the market prices are not available, plot two curves and note the omission in the limitations.

TODO: The TTF return series studied in this thesis is positively skewed instead, as reported in Sec. ???. Supply disruptions such as those described in Sec. ?? raise prices sharply, while no comparable mechanism lowers them. The skewness determines the tilt of the volatility smile described in Sec. 2.4, and therefore the relative value of calls and puts at equal distance from the current price. cite Geman????

6 Discussion

TODO: Limitations: a two-return construction was possible from opening and settlement prices but not pursued — returns would need aggregating back to daily for the Monte Carlo, and would mix a traded price with an exchange mark.

TODO: Since each condition appears once in the data with a single realized target, the empirical conditional distribution is a point mass. Vuletić et al. (2024) note that forecasts close to the target therefore receive high scores from the discriminator, which encourages mode collapse in a classical GAN setup

7 Conclusion and Outlook

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Documentation of AI Tool Usage

Declaration of Authorship

Hereby, I declare that I have independently prepared the examination performance or thesis in the above-mentioned module – for group work, the portion of the work marked with my contribution – and have not used any sources or aids other than those indicated. Texts, ideas, concepts, graphics, etc., taken directly or indirectly from external sources in my statements have been clearly identified as such and provided with full references to the respective source.

All quotations, as well as all sections written or modified using AI-based tools, have been checked for correctness and marked accordingly. Documentation of the AI tools used, along with their purpose of use, is attached to the work. I take responsibility for the quality and all contents of the present work, and my creative influence predominates.

Furthermore, I assure that I have exclusively used AI-supported tools that the examiner explicitly identified as approved aids. The work has not been submitted in the same or a similar form in any other examination.

Fulda, August 29, 2026

Yeva Galstyan